Majumdaar Hosiery ERP · how it works

One-page reference to the logic behind every screen: how a document moves stock and money, how lot costs are built up, how GST, accounts, roles and reports are calculated, and where the data lives.

`Node.js + browser` `no add-ons to install` `SQLite · erp.sqlite` `Hostinger Web App` `4 locations` `9 roles` `18 reports`

PROCESS FLOW · ONE LOT FROM YARN TO CASH

Each step is a document. Saving it moves stock between the four locations and carries the lot's cost forward.

Step 1 — Purchase
`PO → GRN → PB → PAY`
Order yarn, dyes & chemicals, packing. GRN brings stock in at the PO rate and gives each yarn / fabric line its **lot number** (`L-0001`). Bill adds GST and creates the payable.
`Supplier → Main Store`

Step 2 — Yarn issue
`TRF · Yarn Issue`
Store sends yarn (by its lot) to the knitting unit. Cost moves with it.
`Main Store → Knitting Factory`

Step 3 — Knitting
`KNT · keeps lot L-0001`
Yarn consumed → grey fabric under the **same lot number** (kg, rolls, GSM), sent **straight to the colour factory**: no transfer entry. Wastage and knitting labour recorded.
`Knitting → Color Factory`

Step 4 — Dyeing
`DYE · recipe (BOM)`
Grey lot + dyes & chemicals (from Main Store) → coloured lot. Colour, shade, wastage, dyeing labour.
`at Color Factory`

Step 5 — Finishing & packing
`FIN · overhead added`
Coloured lot + packing material → finished rolls (kg, rolls, meters). Month's overhead absorbed here.
`Color Factory → FG Store`

Step 6 — Sales
`SO → DC → INV`
Order at the customer's price; challan dispatches the lot; tax invoice raises the receivable.
`FG Store → Customer`

Step 7 — Collection
`RCT`
Receipt reduces the customer's balance; oldest dues are settled first for ageing.
`Customer → Bank / cash`

LOC Locations & lots

CODE	LOCATION	HOLDS
`STORE`	Main Store	Yarn, dyes & chemicals, packing
`KNIT`	Knitting Factory	Issued yarn (grey fabric only if held back)
`DYE`	Color Factory	Grey waiting to dye, dyed waiting to finish
`FG`	Finished Goods Store	Packed finished rolls

- These four are built in and cannot be deleted; extra stores or job-work locations can be added.
- **Stock is kept per item + location + lot.** The same item in two lots is two separate stock lines.
- **One lot number from purchase to sale.** The GRN gives each yarn or fabric line the next free number (`L-0001` , `L-0002` …); opening stock gets one the same way. Nobody types it. Knitting keeps the lot of the main yarn (the yarn line with the most kg), dyeing and finishing keep the lot they take in, and the challan / invoice sells from it. Dyes, chemicals and packing have no lot.
- **Knitting output** goes to the Color Factory by default ("Grey fabric goes to"). Choosing Knitting Factory holds it there until a *Transfer to Color Factory*; entries made before this option keep their grey at the Knitting Factory.
- **Finished rolls are weighed one by one** at finishing (`20.2 + 50.3 + 39.5` → 110 kg, 3 rolls). A sale takes whole rolls in roll order and cuts the last (23 kg → roll 1 + 2.8 kg of roll 2, leaving 47.5 + 39.5), or the rolls picked on the challan. Stock shows the kg left on each roll.
- Issues and consumptions pick the lot from a list that shows the quantity available there. One yarn lot knitted into two fabrics (e.g. single jersey and rib) keeps one number: stock and colour are kept per item + lot.

DOC What each document does

DOCUMENT	STOCK	MONEY
Purchase order `PO`	—	— (pending tracked)
Goods receipt `GRN`	+ at chosen location, qty × rate	—
Purchase bill `PB`	—	Supplier credited (payable), input GST
Supplier payment `PAY`	—	Supplier debited
Transfer `TRF`	– from, + to (same lot, same cost)	—
Opening / adjustment `ADJ`	+ qty at rate, or – qty (write-off)	—
Knitting `KNT`	– yarn at KNIT, + grey lot at **DYE** (or KNIT if held back)	Labour into lot cost
Dyeing `DYE`	– grey lot at DYE, – chemicals at STORE, + coloured lot at DYE	Chemicals + labour into lot
Finishing `FIN`	– coloured lot at DYE, – packing at STORE, + finished lot at FG (each roll weighed)	Packing + labour + overhead into lot
Overhead `OH`	—	Spread over the month's finished kg
Sales order `SO`	—	— (credit check, pending tracked)
Delivery challan `DC`	– finished lot at FG, roll by roll	Fixes cost of goods sold
Invoice `INV`	– at FG only if no challan	Customer debited (receivable), output GST
Receipt `RCT`	—	Customer credited
Labour payment `LP`	—	Reduces labour payable for its process

LINK Documents that fill from another

- **PO → GRN:** loads only the **pending** quantity per item (ordered – already received against that PO).
- **GRN → Purchase bill:** copies all lines with each item's GST %. A GRN can be billed only once.
- **SO → Delivery challan:** loads pending quantity (ordered – already dispatched); you pick the lot.
- **Challan → Invoice:** copies items, lots, kg and rolls. A challan can be invoiced only once.
- **Rates:** sales lines take the customer's price-list rate (latest "effective from"), else the item's standard rate. Purchase lines take the item's standard rate. GST % always comes from the item.
- **Recipe (BOM):** picking a recipe in dyeing or finishing fills the materials as `qty per kg × input kg` , and rescales if the input kg changes (until you edit a line).
- **Lot pick:** in dyeing/finishing, choosing the lot fills the input kg with what is available; the output keeps the same lot number.

STOCK Stock engine

- Only documents are saved. **Stock, ledger, lot cost, wastage and balances are recalculated** by replaying every document in date order (same date: order of entry). So editing or deleting an old entry corrects everything after it.
- **Valuation** (Settings): **Weighted average** (default) or **FIFO** per stock line. Production lots are valued at their exact built-up cost.
- **No negative stock.** Every save, edit and delete is test-replayed first; if it would make any item/location/lot go below zero, it is refused with the item, lot and shortfall (e.g. *"short by 340"*). An admin can switch this off in Settings; shortages then show in *Stock Exceptions*.
- Rolls are counted alongside kg; meters are recorded at finishing.
- Stock reports can be run "as on" any past date, lot-wise or item-wise, with value and re-order alerts.

COST Exact lot costing

Every kg carries its cost split into **buckets**: yarn · dyes & chemicals · packing · other material · knitting labour · dyeing labour · finishing labour · overhead · wastage. The buckets travel with the fabric, so a finished lot knows exactly what it cost and why.

```
At each stage:
w            = wastage kg ÷ input kg
wastage cost = w × input cost − scrap sale value
output cost  = input cost × (1 − w)
             + wastage cost  (kept in the wastage bucket)
             + materials added + labour (+ overhead)
cost per kg  = total ÷ finished kg   (also per roll, per meter)
```

Worked example (from the test run): 600 kg yarn @ ₹250 = 1,50,000 + chemicals 8,190 + packing 120 + labour 4,680 + 8,775 + 3,360 + overhead 11,200 = **₹1,86,325 for 560 kg = ₹332.72/kg**. Wastage is part of the total, never added twice.

PROD Wastage, labour & overhead rules

- **Wastage** fills in as `input − output` until you type your own. If you enter less, the rest is **process loss** and its cost stays in the good fabric.
- **Labour** fills in as `rate × kg` (rate from the Process master): knitting and finishing on **output** kg, dyeing on **input** kg. It can be overwritten with a lump sum.
- **Overheads** (factory, power/water, machine running, maintenance, other) are totalled per month and charged to lots finished that month at `month's overheads ÷ month's finished kg` . A month with no finishing shows as **unabsorbed** in the Overhead Absorption report.
- Lot tracking starts at the GRN (supplier, kg received) and follows the lot to sales. Yield = finished kg ÷ yarn kg knitted.

P&L Profitability

```
profit = sale value − cost of goods sold
sale value = line amount − its share of invoice discount (ex-GST)
COGS       = exact lot cost of the kg dispatched
```

Cost is taken from the challan (or the invoice when there is no challan). Shown per invoice line, or grouped by customer, item or lot. Dashboard gross profit uses the same calculation.

GST GST logic

- **Rate and HSN per item**, filled into every line; editable per line.
- **Place of supply** = the invoice's own field, else the party's state. Same state as the company → **CGST + SGST** (half each); different → **IGST**.
- **Freight & other charges** on the invoice are part of the taxable value, spread over the lines by value and taxed at each line's rate. A **discount** reduces taxable value **before** GST.
- Grand total rounded to the rupee (round-off shown).
- **States** come from a fixed list with GST codes (e.g. West Bengal 19). **GSTIN** must be 15 characters in the official pattern and its first two digits must match the state; blank = unregistered.
- **Tax invoice print:** both GSTINs with state codes, HSN per line, kg and rolls with each roll's weight, place of supply, HSN-wise tax summary, amount in words, bank details, terms, signature, and Original / Duplicate / Triplicate copy label.
- **GST Summary** report: month-wise output tax (sales) – input tax (purchase bills) = net payable or credit.

ACC Accounts, dues & credit

ENTRY	PARTY LEDGER
Opening balance	Dr or Cr as set on the party
Sales invoice	Dr (customer owes us)
Receipt	Cr
Purchase bill	Cr (we owe supplier)
Payment	Dr

- **Balance** = Dr – Cr. Positive = receivable, negative = payable.
- **Ageing** 0–30 / 31–60 / 61–90 / 90+ days assumes payments settle the **oldest** bills first.
- **Credit control:** saving a sales order or invoice that takes a customer above the credit limit asks for confirmation (warning, not a block). Outstanding report flags *Over limit* and *Overdue* (oldest due older than credit days).
- Payment and receipt screens show the party's current balance before you enter the amount.
- **Labour payable** per process (knitting, color, finishing & packing) = labour on the production entries – labour payments. The payment form shows what is still owed; Settings can set the date it counts from (earlier labour treated as settled).

MST Masters

Items	Code, name, category (Yarn · Dyes & Chemicals · Packing Material · Others · Grey Fabric · Colored Fabric · Finished Goods), unit, HSN, GST %, standard rate, re-order level. Names and codes must be unique.
Parties	Supplier · Customer · Job Worker · Both; state, GSTIN, credit limit & days, opening balance. Customer Master is the same list filtered.
Units	Kg, Nos, Roll, Mtr, Ltr, Pcs, Box (+ your own).
Machines	Code, type (knitting/dyeing/finishing), dia, gauge, feeders, capacity, status; machine-wise output on the knitting page.
Processes	Stage + labour rate ₹/kg used to fill labour.
BOM / Recipes	Dyeing recipes and packing standards as qty per kg.
Price list	Customer + item + rate + effective-from date.

Dropdowns only offer sensible items: knitting shows yarn, dyeing shows grey fabric and dyes & chemicals, sales show finished goods, and so on.

USER Users, roles & sign-in

- **Administrator** has everything, including users, roles, settings, backup and restore.
- Every other login has a **role**. A role sets one level for each of **24 areas**: `No access` hidden and blocked · **View** open & print · **Add & edit** · **Full** also delete.
- **The server enforces it**, not just the menu: typing a blocked screen's address shows *No access*, and lists only contain records the role may see.
- Masters are readable by everyone (needed for dropdowns); changing them needs the Masters area. Transfers split by purpose: grey transfer belongs to Knitting, yarn issue to Store.
- Role edits apply to everyone with that role at once.
- **Sign-in is per browser tab**: reload keeps you signed in as yourself; a new tab asks to sign in; closing the tab signs out. Sessions end after 12 hours idle. They are kept on disk (as a hash, never the token), so a restart or update signs nobody out.
- Passwords: minimum 6 characters, stored scrambled (scrypt), never shown. Usernames have no spaces; capitals don't matter. Inactive users cannot sign in. Admin cannot delete or demote themselves.
- **Sign-in protection:** 5 wrong passwords for one user from one address (or 30 from one address) pause sign-in there for 15 minutes. The server warns while the admin password is still the default.

ROLE Ready-made roles

ROLE	MAINLY WORKS ON
Management	Everything except users & roles
Purchase	Masters, POs, bills; sees GRN & payments
Store / Inventory	GRN, yarn issue, opening stock; sees POs
Knitting	Knitting (grey goes to the colour factory by itself); grey transfer for held fabric
Dyeing / Color	Dyeing
Finishing & Packing	Finishing
Sales	Customers & masters, prices, SO, challans, invoices; sees receipts
Finance & Accounts	Bills, payments, receipts, labour payments, overheads, MIS dashboard
Data Entry	Add & edit in every module, no delete; no costing or accounts reports

All roles are editable, and new ones can be created.

MIS Dashboards

- **Business overview** (roles with the MIS dashboard): sales with gross profit trend, receivables by age, payables, six tiles with trend lines, material in process at each stage, sales vs purchases, production flow with % yield, top customers, cost per kg by lot, stock value, wastage by stage, best sellers, re-order alerts, recent activity.
- **Department analytics** (everyone else): only their own area. Knitting/dyeing/finishing see kg, %, lots, machines, colours, chemicals and waiting lots, **never money**. Store, purchase and sales see their own figures.
- **Period buttons:** this month, last month, 3 / 6 months, this FY (from 1 April), 12 months, or custom dates. Every tile compares with the **previous period of equal length** (▲▼ green when good, red when bad).
- Ranges up to 62 days are charted by day (trend lines as a running total); longer ranges by month. Every chart has a table view and links to its report.

RPT Reports

Inventory: Stock Summary · Stock Ledger · Stock Exceptions. **Purchase:** Purchase register + item-wise · Pending POs. **Production:** Production · Batch/Lot Tracking · Wastage. **Costing:** Lot Costing Sheet · Item-wise Costing · Overhead Absorption · Profitability. **Sales:** Sales register + item-wise · Pending SOs. **Accounts:** Outstanding (receivable/payable) · Party Ledger · Labour Payable · GST Summary.

All filterable (dates, party, item, location, stage), exportable to CSV and printable. Clicking a row opens the document behind it. Each report group is a separate permission.

RULE Checks on save

- Date and required fields; at least one line; quantities above zero (adjustments may be negative, not zero).
- Lot numbers are given only by the ERP (a typed number on a new GRN line is replaced); knitting output goes only to the Color Factory or stays at the Knitting Factory; transfer "from" and "to" must differ; output kg required for production. Roll weights must be numbers; rolls picked must add up to the line's kg and still be in stock.
- Unique names: items, parties, units, locations, machine codes, processes, recipes, usernames.
- **Delete is blocked** when a record is used elsewhere (e.g. an item on a PO, a GRN already billed) or when deleting would break stock later.
- Edits keep the document number; numbers are only issued once a save succeeds.
- Every entry records who created and last edited it, and when.

NO. Numbering

`PO · GRN · PB · PAY · TRF · ADJ · KNT · DYE · FIN · OH · SO · DC · INV · RCT · LP`

Each series runs continuously with four digits (`INV-0001` , `INV-0002` …). Lots run `L-0001` upwards, given at the GRN.

UI Working on the screens

- **Dark theme** by default; sun/moon button switches to light (remembered per computer). Printed documents and reports are always white.
- **Full screen** starts when you sign in; **Esc** returns to normal; the ⛶ button re-enters.
- **Guidance panel** on every entry screen: what is missing (yarn not issued, no grey lot waiting, chemicals or packing short, item in the wrong category or store), where the stock is instead, the step that fixes it (button, or which department to ask) and what comes next. Short lines get a note underneath; stock errors on save say what to do.
- **Ctrl + S** saves a form; Save & new; Print on saved documents. **Enter** opens a highlighted list row.
- Each menu box has its own colour and the screen it opens uses the same colour.
- Works on a phone or tablet through the online address.

DATA Data, backup & safety

- All data is in one **SQLite database**, `erp.sqlite` (SQLite is built into Node.js, nothing to install). One table per area (items, parties, invoices, knitting …) with readable number, date and name columns.
- Each save writes only the changed rows **in one transaction**: stored completely or not at all, so a power cut cannot leave a half-saved entry. If the disk refuses SQLite's fast mode, the ERP switches to the simpler mode by itself.
- **Online:** a Hostinger Business Node.js Web App deployed from GitHub. The database lives in its own folder set by `DATA_DIR` (e.g. `/home/<u-number>/erp-data`), outside the app folder that Hostinger replaces on every deploy. Steps in `DEPLOY.md` .
- **Backups:** the server saves a copy **once a day** by itself into `backups/` beside the database, keeping the newest 30. Settings → Download backup (JSON, restorable) or Download database (SQLite file) for an office copy. **Restore** first saves the current database as `erp-before-restore-….sqlite` .
- **Startup log:** every start and any error is written to `server.log` beside the database, readable in File Manager.
- **Demo data** (12 months, 24 lots, 5 customers) loads only into an empty database.
- **Updates:** push to GitHub and redeploy; sign-ins are kept.
- **Sleep on Hostinger:** an unused app is stopped and starts again on the next visit, so that first page takes a few seconds (a *Waiting for the server…* note shows; reads retry by themselves). A 5-minute ping of `/api/health` keeps it awake.
- **Tests:** `npm test` runs the storage test, the API checks, the two-tab sign-in check and a 110-step department walkthrough on a temporary database.

NEXT Not built yet

- Credit & debit notes (sales returns, rate differences)
- GSTR-1 / GSTR-3B ready reports; e-way bill details; e-invoice (IRN/QR, needed above ₹5 crore turnover)
- Job-work challans & ITC-04; ITC rules (2B matching, blocked credit, reverse charge)
- Invoice series restarting each financial year
- Tally-style shortcut keys and Alt + G search (proposed)
- Knitting recipes (fixed yarn blends) (proposed)

GST rates and HSN codes are set per item. Have your CA confirm them.

Figures in the costing example come from the automated end-to-end test.